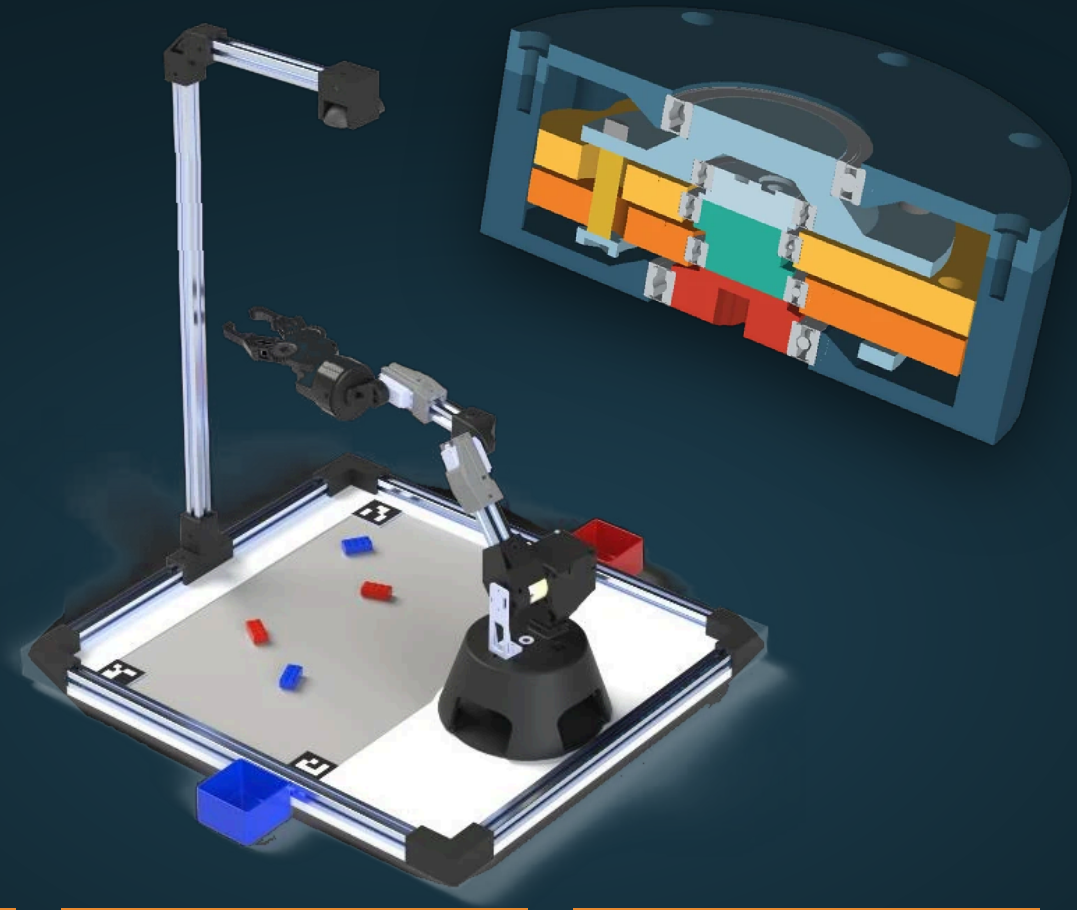
ENGINEERING PORTFOLIO

Jamal Charkie

Mechanical engineering student at DTU (BEng, 5th semester) and teaching assistant, focused on mechanical design and mechatronics.

I like taking an idea all the way: **calculating it, designing it in CAD and building a prototype** that gets tested and used.



OPEN TO

An engineering internship from February 2027, and a student job alongside my studies.

01 · DTU

ARES-4 modular robot arm

48 / 50

bricks sorted, fully automatic

02 · PERSONAL PROJECT

Cycloidal QDD actuator

15 : 1

design ratio · in progress, CAD complete

03 · CLIENT PROJECT

Wall-mounted QR scanner

V2

in use every week

04 · DTU

Motor-driven syringe pump

99–101 %

of the set flow

Also built: a vision-guided nerf turret (YOLO, steppers) · a bike-powered phone charger · a table football goalkeeper bot.

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3D models and video: **jamalch15.github.io/portfolio**

01

COMPLETED DTU · team of 3

ARES-4: a modular robot arm, built for many tasks

Its tools change by hand with a twist, and a browser dashboard sets up and runs new jobs. Tested on **fully automatic LEGO sorting**: find a brick, pick it, drop it in its colour's bin, repeat until the table is clear.

SKILLS **Robot kinematics** · **Modular mechanism design** · Embedded firmware

48 / 50
bricks sorted with no one touching it

≈500 g
payload carried

≈7 Nm
shoulder peak, for ≈4 Nm needed

WHAT I DID

In a team of three (DTU, 2026), I drove the **mechanical design, the electronics and the software**: motor sizing and CAD, KiCad wiring, ESP32-S3 firmware, kinematics and the control dashboard.

DESIGN DECISIONS

Strong enough at full reach. About 4 Nm needed at the shoulder; a 27:1 geared stepper gives about 7 Nm peak, confirmed by our torque test.

Light up top. Servos at the elbow and wrist keep the mass near the base.

Quick-swap tools. Gripper or electromagnet, changed by hand without tools; a sideways pogo connector powers the tool when it locks.

Encoder-corrected shoulder. An AS5048A measures the real angle after each move.

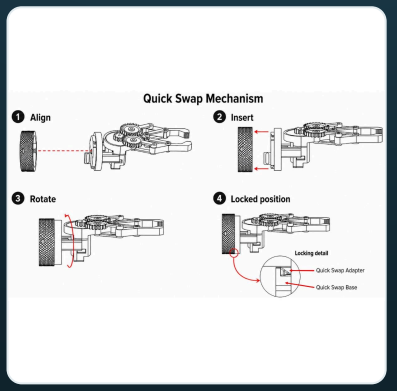
Known limits. Backlash limits precision; next step: closed-loop control on every joint.



THE ARM IN 3D, FROM THE ORIGINAL SOLIDWORKS ASSEMBLY



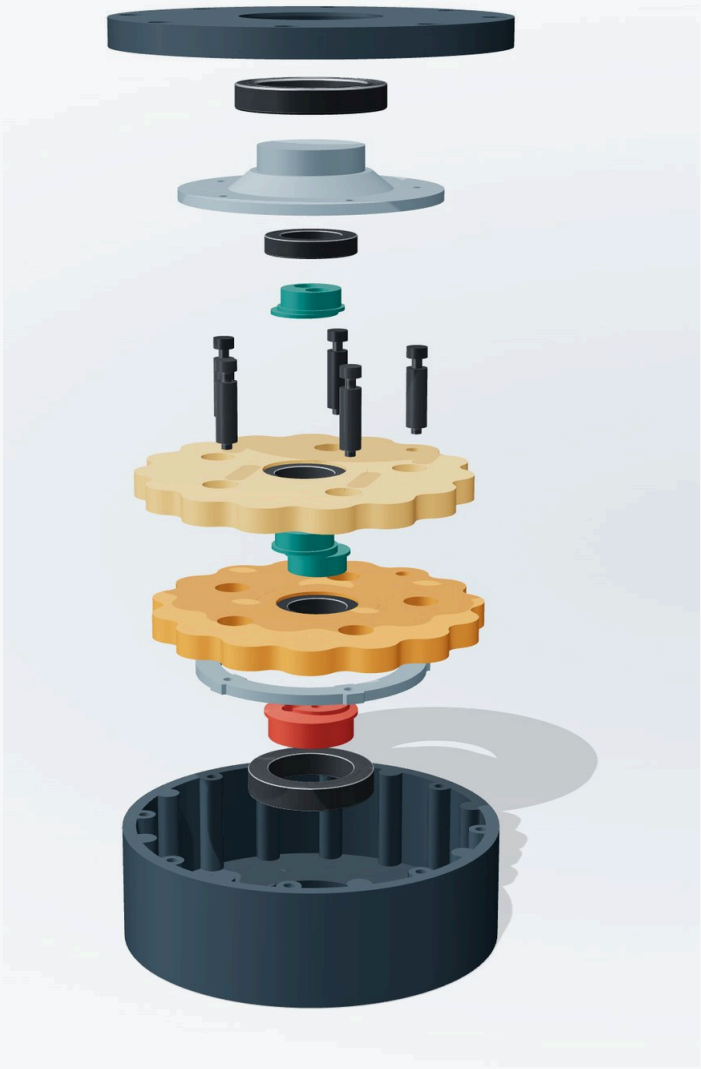
THE REAL CELL



QUICK-SWAP: ALIGN, INSERT, ROTATE, LOCK



DASHBOARD: GHOST ARM AND PLANNED PATH



02

IN PROGRESS · CAD COMPLETE Independent project

A printable robot joint, designed to be backdrivable

A low-ratio cycloidal drive on a drone motor: the quasi-direct-drive kind used in legged robots, **designed to be printed at home** around a few bought parts.

SKILLS Gear reduction design · Iterative prototyping · Design for 3D printing

15 : 1
reduction, in a 12–16:1 target

≈16 Nm
peak torque, design target

≈470 rpm
top speed, design target

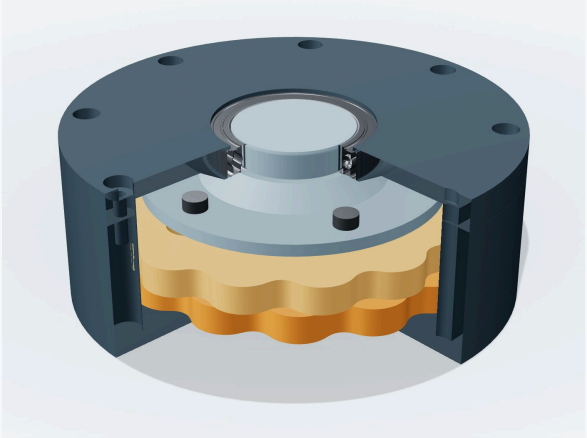
DESIGN DECISIONS

Low ratio on purpose. The aim: a joint that stays backdrivable and quick to respond.

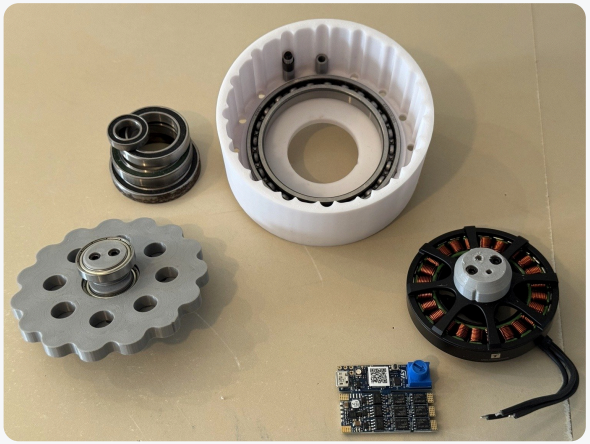
Drone motor, not a stepper. Sized for a T-Motor MN6007 II outrunner, with field-oriented control planned.

No ring pins. The discs roll on a profile in the printed housing; printed sleeves over screws had caused seam friction.

Serviceable shaft. A three-piece eccentric shaft clamped with M3 screws, so the bearings fit inside the discs.



CUTAWAY: TWO DISCS INSIDE THE LOBED HOUSING



FIRST PROTOTYPE: PRINTED AND FITTED WITH REAL BEARINGS

Next: fit samples, assembly and hand-turn tests of this design.

03

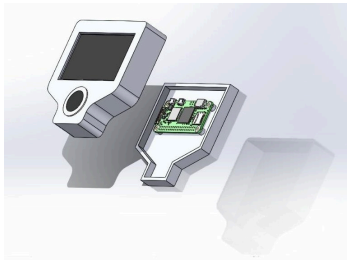
IN USE EVERY WEEK · Independent, for an association

QR check-in terminal: a year in service, then a redesign

I designed and built the hardware: the enclosure and wall mount around the display, GM60 scanner and Raspberry Pi. The app software was handled separately.

SKILLS · **User-centred design** · **Redesign from field use** · Enclosure and mount design

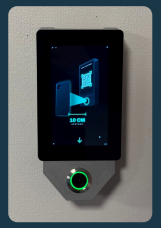
V1 · A YEAR IN SERVICE



Issues and real use **logged for a year**; then the association asked for an improved unit.

→

V2 · REDESIGN, IN SERVICE



- Larger official 5-inch Raspberry Pi display
- A clearly separate scan opening
- An easier-to-read front
- Wiring through the wall: no visible cables
- Easier, faster, more robust start-up
- Quick to assemble and service

In use at the association every week.

DESIGN DECISIONS

Designed around the real parts. The display, scanner and Raspberry Pi set the packaging; the CAD covers fasteners, clearances, cable routing and assembly access.

Print-ready. Every housing part is designed for FDM, then printed, assembled and tested on the real hardware.

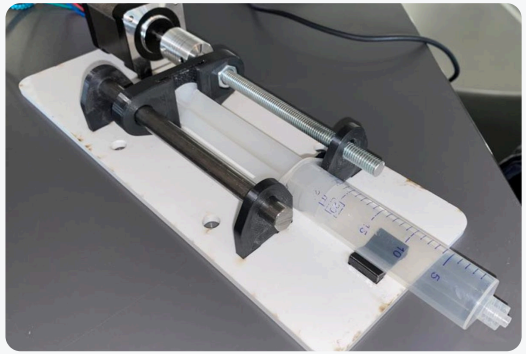
04

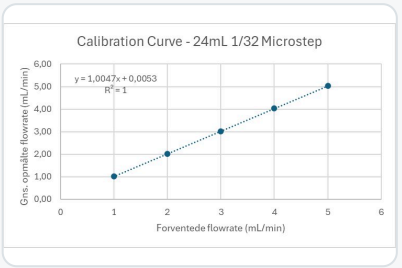
COMPLETED · DTU course project · team of 3

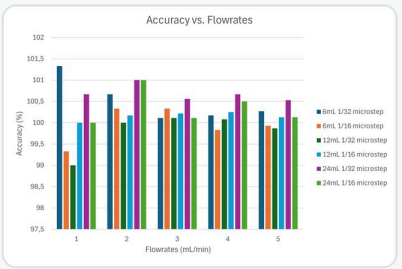
A printed syringe pump within 99–101 % of set flow

A modular pump that pushes and pulls, takes 6–24 mL syringes and turns stepper rotation into precise plunger travel. **I modelled it, built it and measured it.**

SKILLS · **Calibration and validation** · **Linear drive design** · Rapid prototyping







99–101 %
of set flow, 6, 12 and 24 mL syringes

1–5
mL/min tested, 3 repeats per point

0.01 g
scale for every measurement

WHAT I DID

CAD modelling (with feedback from the group), **building the prototype**, and the **data collection and processing**.